Machine Learning Engineer Assessment Report

Freight Rate Prediction

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This report summarizes data exploration, data cleaning, validation strategy, modeling experiments, feature engineering, and findings from the freight rate prediction assessment.

1. Objective and Dataset

The objective was to train a regression model to predict the posted freight rate (posted_rate) for each load in the supplied validation dataset. The development data contained load attributes such as pickup and delivery locations, geographic coordinates, distance, equipment type, weight, date, market_index, quote_signal, and the target posted_rate. The validation file contains 12,000 loads requiring predictions.

The load identifier (load_id) was treated as an identifier rather than a predictive feature. The target variable was posted_rate.

2. Data Exploration and Quality Checks

Initial descriptive statistics indicated that distance and posted_rate have right-tailed distributions. The target had a mean of approximately 2,374 and a median of approximately 2,031, with a maximum around 25,533. This suggests that a relatively small number of high-rate observations may contribute substantially to RMSE.

Data Cleaning

The following data-cleaning steps were performed:

- **Missing values:** Missing numerical values were filled using the median of the corresponding feature. Median imputation was selected because the median is less sensitive to extreme values than the mean.
- **Outlier detection and removal:** Outliers were detected, and the identified records were removed. The removed records represented approximately **7.35% of the dataset**.
- **Duplicate check:** The dataset was checked for duplicated rows, and **no duplicated rows were found**.

The weight column contained missing values and negative values, including a minimum near -47,500. Negative weights require careful interpretation against the dataset specification to determine whether they represent invalid records or another encoding.

CatBoost does not require normally distributed input features. Therefore, transformations were evaluated based on validation performance rather than an objective of forcing the features or target to follow a normal distribution.

For a production-quality evaluation, imputation values and outlier thresholds should be calculated using the training data only and then applied consistently to the relevant validation data to prevent data leakage.

3. Validation Strategy

A time-aware expanding-window validation strategy was used to evaluate how well the model could predict future freight rates. This approach preserves chronological order and avoids randomly mixing earlier and later observations.

Two temporal validation approaches were considered.

3.1 Monthly Expanding-Window Validation

The monthly approach uses earlier months for training and the next month for testing. After each test period, the training window expands to include the newly available earlier data.

Fold	Training Data	**Testing Data**
1	Months 1	Months 2
2	Months 1-2	Months 3
3	Months 1-2-3	Months 4
4	Months 1-2-3-4	Months 5

This process continues chronologically for the available months.

Why this approach was used: It simulates a realistic forecasting scenario in which historical data is available for training, while the model must predict freight rates in a future month.

3.2 Daily Expanding-Window Validation

The daily approach follows the same principle but evaluates performance at a daily level.

For each validation day, the model is trained using all eligible data from earlier days and evaluated on the next day.

For example:

Fold	Training Data	**Testing Data**
1	Day 1	Day 2
2	Day 1-2	Day 3
3	Day 1-2-3	Day 4
4	Day 1-2-3-4	Day 5

The process continues, with the training dataset expanding as more historical observations become available.

The quick daily experiments used the first 30 available dates, yielding 29 daily validation steps. This allowed practical comparisons of modeling choices while preserving temporal order.

The reported RMSE and MAE results below are approximate averages from the daily temporal validation experiments. A separate monthly performance score should only be reported after calculating the metrics across the monthly test folds.

4. Baseline Model and Results

CatBoostRegressor was selected as the primary model because the dataset combines numerical variables with categorical variables such as pickup, delivery, and equipment. CatBoost can handle categorical features directly and can learn nonlinear relationships and interactions.

Metric	Approx. result	Interpretation
RMSE	256	Penalizes large errors heavily; a few extreme misses have a strong effect.
MAE	100	Average absolute prediction error is approximately 100 rate units.

These are approximate results from the daily temporal validation experiments and should be interpreted as experimental estimates, not as a guarantee of performance on the final hidden validation set.

5. Feature Engineering and Experiments

Experiment	Validation result	Conclusion
Baseline CatBoost	Approx. RMSE 256; MAE 100	Used as the reference model.
Calendar features (day of week, day of month, month)	No meaningful improvement	Retained only if useful for the final pipeline; not a demonstrated performance gain.
Route feature (pickup → delivery)	No meaningful improvement	Added as a categorical representation of the route.
Coordinate differences (latitude/longitude deltas)	No meaningful improvement	Tested as directional/geographic features.
Weight per mile and distance scaling	No meaningful improvement	Tested simple interactions involving weight and distance.
Historical route mean rate	Avg. RMSE approx. 257.17; MAE approx. 98.50	Slight MAE improvement but no RMSE improvement; limited value for the current objective.
Log distance	Avg. RMSE approx. 257.31; MAE approx. 100.57	Slightly worse than baseline.
Log-transformed target (log1p)	Avg. RMSE approx. 272.43; MAE approx. 105.89	Worse than baseline after reversing the transformation.
Distance-only polynomial models	RMSE approx. 298	Underperformed CatBoost, indicating that distance alone is insufficient.

6. Feature Importance and Observations

CatBoost feature importance indicated that distance was the dominant feature (approximately 88.8% of total importance in the reported run), followed by equipment type (approximately 4.8%) and weight (approximately 1.7%). Market_index and quote_signal contributed smaller amounts. Geographic coordinates had comparatively low importance in that run.

Exploratory analysis showed that rate per mile generally decreases as trip distance increases, while equipment type affects typical rate per mile. However, these variables did not fully explain the largest prediction errors. The rate_per_mile variable was used for analysis only and was not used as a model input because it is calculated from the target and would cause target leakage.

7. Error Analysis

The largest errors included cases where the actual posted rate was several thousand units higher than the model prediction, as well as some cases where the model substantially overpredicted the actual rate. For example, several loads with actual rates around 5,000–6,400 were predicted around 1,000–1,900, while some long-distance loads had predictions much higher than their actual rates. These cases indicate that the model captures the dominant distance relationship but does not explain every exceptional price.

Potential next steps are to inspect the worst 20–50 errors across all validation days, check whether those records share equipment, route, date, weight, market, or data-quality patterns, and confirm whether the extreme target values are valid business observations. Additional features should only be added when supported by this analysis.

8. Conclusions and Recommended Next Steps

- CatBoost was the strongest tested approach among the experiments documented here, with approximate temporal validation performance of RMSE 256 and MAE 100.
- Additional simple feature engineering and target/distance log transformations did not improve the overall result meaningfully.
- RMSE is strongly affected by a small number of large prediction errors; error analysis is therefore a higher-value next step than continuing arbitrary feature additions.
- Investigate negative and missing weight values using the dataset specification before applying cleaning rules.
- Before submission, train the chosen pipeline on all eligible development data, generate predictions for every row in validation.csv, and verify that the output contains exactly 12,000 rows, all required load_id values, and no missing or invalid predictions.
- If time permits, run a broader temporal backtest across more dates to check that performance is stable over time.

9. Scope and Limitations

This report documents the experiments completed and the approximate results available during development. Some feature experiments showed little or no improvement, and the quick daily temporal validation run was limited to the first 30 available dates.

The final validation-set performance is unknown until predictions are evaluated against ground truth, which may not be available.

The RMSE and MAE figures reported in this document refer to the daily temporal validation experiments; they should not be interpreted as monthly validation results unless those results are calculated separately.

The 7.35% outlier-removal figure describes the data-cleaning step. The outlier detection method, thresholds, and treatment of extreme target values should be documented in the implementation to ensure reproducibility and avoid removing legitimate high-value freight loads.

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